

BINUS University

Undergraduate Program <i>Regular/Smart Program/Global Class*)</i>		Term : Odd/Even/Short *)
☐ Mid Exam ☐ Final Exam ☐ Short Term Exam ☒ Others Exam : Mid Exam PPT1 7 MP2		Academic Year : 2019 / 2020
Faculty / Dept. : School of Computer Science		Student ID :
Code / Course : COMP6047 – Algorithm And Programming		
Class : LAT8 (111AT)		N a m e :
Shift : Morning		
Day/ Date : <i>Wednesday</i> , August 28 th 2019		Signature :
Time : 08:30 – 10:10 (..... Minutes)		
Lecturer : D5797 – Felix Jingga, S.Kom.		
Exam Type : Open Book/Close Book/Open e-Book/ Submit Project/Oral Test *)		
Equipment : Exam Booklet/Calculator/Dictionary/ Laptop/Tablet/Smartphone/ Drawing Paper – A3/Drawing Paper – A2 *)		
*) <i>Strikethrough the unnecessary items</i>		
<i>Please insert the test paper into the exam booklet and submit both papers after the test!!!</i> <i>The penalty for CHEATING is DROP OUT!!!</i>		

Rules:

Students **ARE NOT ALLOWED** to bring:

- Any digital materials (e.g., e-books, softcopy codes, etc.)
- Any digital devices (e.g., flash disks, external hdd, **apple watch or any other smart watches**, etc.)
- Any other electronic devices (e.g., smartphone, handphone, calculator, laptop, etc.)
- Any communication devices should be turned off during this exam.

Fair Examination

A student may be zero-scored for any cheat or attempt of cheat (submitting a solution which is not made by yourself is considered as a very serious cheating) and will be processed through University's policy regarding cheating. A student may also be excused from the exam room for any activity that jeopardized the exam (e.g., **hacking the exam system**, dislodging extension cords, distracting behavior, etc.). All students should uphold the honesty and the spirit of fair examination.

Problem A (50%)

Least Common Multiples

Jojo is solving a mathematic problem. Given N numbers (A_i), Jojo needs to calculate the Least Common Multiples of those N numbers. A Least Common Multiples is a number X **that can be divided by all those N numbers**, and X is the **least/lowest number possible**.

For example, the Least Common Multiples of 6 and 8 is 24. Another example, the Least Common Multiples of 15, 45, and 105 is 4725. Note that 4725, 4740, 4755, etc. are the numbers that can be divided by 3, 5, and 15, but 4725 is the **least number from all those numbers** that can be divided by those 3 numbers.

Format Input

The first line of the input consists of a single integer T , denoting the number of test cases. After that, T lines followed. Each of that T lines denotes a single test case that contains an integer N , followed by N numbers $A_1 A_2 A_3 \dots A_N$

Format Output

For each test case, output a single line that denotes the Least Common Multiples of those N numbers.

Constraints

$$1 \leq T \leq 100$$

$$2 \leq N \leq 100$$

$$1 \leq A_i \leq 1000$$

The Least Common Multiples will not exceed **integer range**.

Solve this problem using the materials that you learn **from the start of the semester up to mid-term**. **DO NOT USE** "shortcut" method such-as built-in function, etc.

Sample Input	Sample Output
5	24
2 6 8	525
3 15 35 75	385
3 5 7 11	1575
3 45 75 105	55476
2 804 828	

Notes

Even though it's not stated explicitly, you should know by now that excessive space / newline is treated as **WRONG ANSWER**.

Least Common Multiples is known in Indonesia as KPK (Kelipatan Persekutuan Terkecil).

The moment when you want to quit is the moment when you need to keep pushing.
– Unknown

Problem B (50%)

Another K-th Digit

Bibi spends all of her time to type. Today, she types numbers from 1 to 10. But, the space button in her keyboard is broken, so each number doesn't have space to separate them. But the way she types the number is unique:

11212312341234512345612345671234567812345678912345678910

Now, this sequence looks interesting for Lili. She wonders if Bibi types the numbers from 1 to N, what is the k-th digit that Bibi types. On the example above, the 11th digit is "1", the 1st digit is "1", and the 18th digit is "3".

Format Input

The first line of the input consists of a single integer T, denoting the number of test cases. After that, T lines followed. Each of that T lines denotes a single test case that contains an integer N and K.

Format Output

For each test case, output a single line containing the K-th digit in the sequence.

Constraints

$$1 \leq T \leq 100$$

$$1 \leq N \leq 300$$

$$1 \leq K \leq 100.000$$

There'll be **no case** when the resulting sequence from 1 to N is **less** than K in length.

Sample Input	Sample Output
3	3
3 6	1
30 1	6
21 21	

Notes

If N = 1, then the unique number is : 1;

If N = 2, then the unique number is: 112;

If N = 3, then the unique number is: 112123;

If N = 4, then the unique number is: 1121231234;

And so on.

In the first sample N=3, Bibi writes:

112123

and the 6th digit is "3".

In the third sample N=21, Bibi writes:

112123123412345123456123456712345678123456789123456789101234567891011..... (still continue)

and the 21st digit is "6"

Even though it's not stated explicitly, you should know by now that excessive space / newline is treated as **WRONG ANSWER**.

What we face may look insurmountable. But I learned something from all those years of training and competing. I learned something from all those sets and reps when I didn't think I could lift another ounce of weight. What I learned is that we are always stronger than we know
– Arnold Schwarzenegger

-- Selamat mengerjakan --